

Cordless Garden Trimmer

Battery Retention Latch

Product context

The company is redesigning the battery retention latch used in a cordless garden trimmer.

The latch retains the removable battery during normal use and must allow the user to release it manually. The product is used outdoors and may be exposed to vibration, drops, rain, soil, fertiliser residue, lubricants, and seasonal temperature changes.

The current latch has received complaints about cracking and occasional battery release during cold-weather use.

Product objectives and priorities

- Prevent accidental battery release during normal operation.
- Improve durability during cold-weather use.
- Maintain easy manual battery removal.
- The expected operating-temperature range is approximately -15 C to 50 C.
- A reduction in battery-release force of approximately 20% would improve usability.
- Reliability has higher priority than release-force reduction.
- The component should contain at least 25% post-consumer recycled material by mass.
- The visible surface should remain black and matte.
- The existing mould and assembly line should preferably be reused.
- Component cost should not increase by more than approximately 5%.
- The geometry around the flexible latch arm is still under review.

Current component and industrial context

Field	Information	Status
Component mass	34 g	Confirmed
Nominal wall thickness	2.2 mm	Confirmed
Manufacturing process	Injection moulding	Confirmed
Current material	Impact-modified ABS; exact grade unknown	Partially known
Current material density	Unknown	Not available
Current material price	Approximately EUR 2.60/kg	Estimate
Declared recycled content	0%	Confirmed
Assembly	Two screws and one snap-fit	Confirmed
Annual volume	Approximately 240,000 parts	Forecast
Current cycle time	29 seconds	Confirmed
Current scrap rate	Approximately 4%	Estimate
Existing mould	Strong preference to reuse it	Stakeholder preference

No standardised material-specimen data is available for the current ABS grade.

Existing evidence comes from field returns, finished components, supplier documents, and informal internal observations.

Field issue and available evidence

Customer returns

- 22 cracked latch components were returned during the previous winter season.
- 17 of the 22 returns came from regions where temperatures had fallen below 5 C.
- The actual operating temperature at the moment of failure is unknown.
- The product age, impact history, and battery-removal frequency are unknown.
- Four customers reported that the battery released unexpectedly.
- Most visible cracks appeared near the root of the flexible latch arm.
- Several returned parts also showed whitening around the snap-fit region.

Internal cold-use observation

- Six complete products were conditioned for three hours at -10 C.
- Operators repeatedly inserted and removed the battery by hand.
- Three of the six latch components cracked.
- No room-temperature control group was tested.
- The insertion and removal forces were not measured.
- Different operators performed the test.
- The number and speed of cycles were not kept consistent.
- The test protocol was not standardised.

Assembly observations

- No immediate cracking is normally observed during assembly.
- Whitening is sometimes visible around the flexible latch arm.
- Screw torque is not systematically recorded.
- Battery insertion force is not recorded.
- Operators sometimes press the battery into place a second time when the first insertion does not produce an audible click.
- Dimensional variation between battery suppliers has not been measured.

Stakeholder notes

Product manager

Reliability is the first priority. The battery must not release accidentally.

The latch should also feel easier to operate because some users complain that the current release force is too high.

The new part must still look black, matte, and robust.

Product engineer

The root cause may involve low-temperature material behaviour, the sharp geometry at the latch-arm root, residual moulding stress, excessive battery insertion force, assembly conditions, or a combination.

No acceptable deformation limit has been defined.

No validated numerical stiffness or strength target exists.

Manufacturing engineer

The current mould and machine were developed for injection-moulded ABS.

Large changes in shrinkage may cause dimensional or assembly problems.

The mould contains a thin flexible section near the latch arm.

Abrasive reinforcements may increase mould wear.

A cycle-time increase above 8% should be avoided.

No validated minimum-flow requirement is available.

Sustainability manager

At least 25% post-consumer recycled material is expected.

Feedstock origin and traceability must be documented.

Recycled content should not be accepted if it significantly reduces durability or causes unacceptable process variability.

Quality manager

The new material must have a tensile strength of at least 140 MPa.

This value was copied from a supplier presentation. No internal analysis has established a relationship between this value and battery-latch performance.

Supplier suggestion

"We recommend a 30% glass-fibre-reinforced PA66. Its tensile strength exceeds 150 MPa, so it will eliminate the cracking and prevent accidental battery release."

Additional information

- The supplier has not tested the proposed grade in the current mould.
- No finished latch components have been produced from the material.
- No low-temperature component-level evidence has been supplied.
- No data has been provided concerning battery-release force.
- The proposed grade's recycled content has not been confirmed.
- No appearance samples have been supplied.
- The supplier has not estimated the effect on cycle time, mould wear, or component cost.

External research note included in the dossier

"Glass-filled PA66 is commonly used in automotive applications and is therefore suitable for high-reliability outdoor products."

The presentation does not identify:

- the original source;
- a specific material grade;
- test methods;
- environmental conditions;
- moisture-conditioning state;
- component geometry;
- or application-specific evidence.

Treat this note according to the level of evidence you consider appropriate.